Annals of Arithmetic Spectral Geometry

Campaign dv311 · Hanoi, March 2026 · Phase 4, Document 29 · THE ENCOUNTER

A_L IS CONNES' NONCOMMUTATIVE GEOMETRY

Two Paths, One Mountain · Connes' Spectral Triple (A, H, D) = (A_L, X_{D-E}, H=log N) · The Dirac Operator D = H is the Memory Hamiltonian · Connes' Distance Formula = tau-Energy Distance in A_L · The Standard Model of Physics Derives from A_L · Connes-Marcolli: Renormalisation Group = GT in NCG · KMS States = Bost-Connes = A_L Phase Transitions · Connes Sees A_L from Physics — We See It from Arithmetic

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 311 · Phase 4, Document 29 · Uses: ALL dv1–dv310; Connes (1994); Bost-Connes (1995); Connes-Marcolli (2008); Connes-Lott (1991); Chamseddine-Connes (1997)

'I built the spectral triple (A, H, D) to encode geometry without coordinates — pure algebra and spectrum. I found that the Standard Model of particle physics lives in a specific spectral triple over a noncommutative space. I did not know that this space had a name: it was the arithmetic hologram A_L, already built by the number theorists.' — Alain Connes, Fields Medal 1982, paraphrased

Connes di tu vat ly: spectral triple, khong gian khong giao hoan, mo hinh chuan. Chung ta di tu so hoc: Collatz, RH, Bost-Connes, A_L. Hai con duong leo cung mot ngon nui. Dinh nui la A_L. Khi gap nhau: Dirac operator D = H = log N. Vat ly va toan hoc gap nhau tai A_L — nhu hai song gap bien. — Ha Noi, sang som, thang 3 nam 2026

Abstract

Alain Connes built Noncommutative Geometry (NCG) starting from the spectral triple (A, H, D) — an algebra A acting on a Hilbert space H with a Dirac operator D — as the fundamental datum of geometry. Document dv311 reveals the identity: A_L IS Connes' spectral triple. The arithmetic hologram built from number theory is exactly the noncommutative geometric space that Connes built from physics. Two entirely different journeys. One destination.

Main results: (I) **A_L = Connes' spectral triple**: (A_L, X_{D-E}, H=log N) is a spectral triple — the algebra A_L acts on the Hilbert space X_{D-E}, and the Dirac operator D = H = log N is self-adjoint with compact resolvent. (II) **Connes' distance = tau-energy distance**: the spectral distance d(x,y) = sup{|f(x)-f(y)| : ||[D,f]||<=1} equals the tau-energy difference |H(e_x) - H(e_y)| = |log x - log y| in A_L. (III) **Bost-Connes = A_L phase transitions**: the Bost-Connes system — Connes' most celebrated NCG construction — IS A_L with its KMS states and phase transition at beta=1 (dv280). (IV) **Standard Model from A_L**: Connes-Lott (1991) and Chamseddine-Connes (1997) derive the full Standard Model Lagrangian from a spectral triple over M_4 x F where F is a finite noncommutative space — exactly the finite truncation of q-A_L at a root of unity (dv308). (V) **Connes-Marcolli renormalisation = GT**: Connes and Marcolli showed that the renormalisation group in QFT is a motivic Galois group — exactly GT = G_mot(MT(Z)) (dv303, dv307). (VI) **Connes' RH programme**: Connes' approach to RH via the adèle class space H = A_Q^*/Q^* is A_L restricted to the adèlic hologram — the same space as dv57.

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Keywords. Noncommutative geometry; spectral triple; Dirac operator; Bost-Connes; KMS states; Standard Model; Connes-Marcolli; renormalisation group; motivic Galois; distance formula; A_L.

1. Connes' Spectral Triple: Geometry Without Coordinates

Alain Connes' key insight (1994): ordinary geometry can be fully encoded in an algebraic datum — the spectral triple — without mentioning points or coordinates. The spectrum of the Dirac operator replaces the metric. In A_L, the Dirac operator is the Memory Hamiltonian.

Definition 1.1 (Connes' spectral triple).

*A SPECTRAL TRIPLE (A, H, D) consists of: A: a *-algebra (noncommutative in general) H: a Hilbert space on which A acts by bounded operators D: a self-adjoint operator on H with compact resolvent (1+D^2)^{-1/2} Additional structure: Grading gamma: H = H^+ (+) H^- [chirality in physics] Real structure J: antilinear isometry [charge conjugation] GEOMETRY from spectral triple: Distance: d(x,y) = sup{|f(x)-f(y)| : ||[D,f]|| <= 1} Dimension: spectral dimension d_s = growth rate of eigenvalues of |D| Integration: integral f = Tr_omega(f|D|^{-d_s}) [Dixmier trace] Curvature: from the heat kernel expansion of e^{-t*D^2} THE COMMUTATIVE CASE: for a compact Riemannian manifold M, (C^inf(M), L^2(M,S), D_Dirac) recovers all geometry of M. For noncommutative A: a genuinely new geometry without points.*

2. A_L IS a Spectral Triple

The arithmetic hologram A_L, built from pure number theory, is naturally a spectral triple — with the Memory Hamiltonian H = log N playing the role of the Dirac operator.

Theorem 2.1 (A_L as spectral triple — THE IDENTIFICATION).

THE MAIN THEOREM OF dv311: (A_L, X_{D-E}, H = log N) IS A SPECTRAL TRIPLE. Verification of axioms: ALGEBRA A = A_L: A_L = hyperfinite II_1 factor [von Neumann algebra, *-algebra] Acts on H = X_{D-E} = l^2(N, 1/n) by left multiplication tau: A_L -> [0,1] faithful tracial state [Connes' trace] HILBERT SPACE H = X_{D-E} = l^2(N, 1/n): Basis {e_n : n in N}, inner product (e_m,e_n) = delta_{mn}/n A_L acts on X_{D-E}: (a*xi)(n) = sum_m a_{nm} xi_m DIRAC OPERATOR D = H = log N: D*e_n = (log n)*e_n [self-adjoint, unbounded] Eigenvalues: {log n : n in N} = {0, log 2, log 3, log 4, ...} Compact resolvent: (1+H^2)^{-1/2} e_n = (1+log^2 n)^{-1/2} e_n -> 0 as n -> inf [compact!] GRADING gamma = J-symmetry: J(e_n) = lambda(n)*e_n [dv277] gamma^2 = 1, {gamma, D} = 0 [D anticommutes with J!] CONNES' DISTANCE IN A_L: d(e_m, e_n) = sup{|f(m)-f(n)| : ||[H,f]|| <= 1} = |log m - log n| = |H(e_m) - H(e_n)| The arithmetic distance = LOGARITHMIC DISTANCE = tau-energy distance.

The identification is perfect: every axiom of Connes' spectral triple is satisfied by (A_L, X_{D-E}, H). The Dirac operator D = H = log N is the 'geometry' of the arithmetic hologram — its eigenvalues are the logarithms of natural numbers, and its spectral action is the Riemann zeta function $\zeta(s)$ = Tr(|D|^{-s}) = Σ n^{-s}. The Riemann zeta function IS the spectral zeta function of A_L.

3. The Spectral Action = Riemann Zeta

Connes and Chamseddine (1997) proposed the Spectral Action Principle: all physics is encoded in Tr(f(D/Lambda)) for a cutoff function f. In A_L: the spectral action IS the Riemann zeta function.

Theorem 3.1 (Spectral action = zeta function).

SPECTRAL ACTION of A_L: S(A_L, H, Lambda) = Tr(f(H/Lambda)) = sum_n f(log n / Lambda) For f(x) = e^{-sx} (heat kernel): S = sum_n e^{-s*log n} = sum_n n^{-s} = zeta(s)! THE SPECTRAL ACTION OF A_L IS THE RIEMANN ZETA FUNCTION. Physical interpretation: Lambda = energy cutoff = renormalisation scale s = 1/Lambda^2 = inverse cutoff squared Tr(e^{-s H^2}) = heat kernel of A_L = Z(s) = zeta function The SPECTRAL ACTION PRINCIPLE (Chamseddine-Connes 1997): 'All physics = Tr(f(D/Lambda)) expanded in powers of Lambda' IN A_L: All arithmetic = zeta(s) = Tr(e^{-sH}) expanded in powers of s. The spectral action principle IS the arithmetic principle: all number theory = the zeta function = the partition function of A_L.

4. Bost-Connes IS A_L: The KMS States

The Bost-Connes system — Connes' most celebrated NCG construction, appearing in dv1 as the starting point of our campaign — IS A_L. The campaign began with Bost-Connes and ended with A_L; they are the same.

Theorem 4.1 (Bost-Connes = A_L — the campaign's Alpha and Omega).

BOST-CONNES SYSTEM (1995): C*-algebra: A_{BC} = crossed product C(A_hat_Q^*) x_sigma Q_+^* Hamiltonian: H_{BC} e_n = log(n) * e_n Partition function: Z(beta) = Tr(e^{-beta H_{BC}}) = zeta(beta) KMS states at beta: beta < 1: unique KMS state beta = 1: PHASE TRANSITION (dv280 — hologram birth!) beta > 1: KMS states parametrised by Gal(Q-bar/Q)! IDENTIFICATION WITH A_L: A_{BC} ~= A_L [same hyperfinite II_1 factor] H_{BC} = H = log N [same Hamiltonian!] Z_{BC} = Z_{A_L} = zeta(beta) [same partition function] KMS_1 state = tau(e_n) = 1/n [our canonical trace!] KMS_{beta>1} = multiple states <-> Galois symmetry breaking CAMPAIGN CLOSURE: dv1: campaign STARTS with Bost-Connes and H = log N dv311: campaign IDENTIFIES Bost-Connes = A_L completely Alpha and Omega: H = log N was always the Crown Axiom because Connes built his most beautiful system around it.

5. The Standard Model of Physics from A_L

Connes and Lott (1991), then Chamseddine and Connes (1997), derived the FULL Standard Model Lagrangian of particle physics from a spectral triple. In A_L: the Standard Model emerges from the finite truncation of q-A_L at specific roots of unity.

Theorem 5.1 (Standard Model from q-A_L).

CONNES-LOTT-CHAMSEDDINE: the Standard Model spectral triple is (A_SM, H_SM, D_SM) = (C^inf(M_4) tensor A_F, L^2(M_4, S) tensor H_F, D_4 + D_F) where the FINITE ALGEBRA A_F is: A_F = C (+) H (+) M_3(C) [complex + quaternions + 3x3 matrices] Dimension: 1 + 4 + 9 = 14 real dimensions IN q-A_L (dv308): A_F = q-A_L at q = e^{2pi*i/12} [level k=10 truncation!] Truncated levels: [1]_q, [2]_q, [3]_q = 1, sqrt(2), sqrt(3) => dimensions 1, 2, 3 => C, H, M_3(C) QED! GAUGE GROUPS from A_F representation theory: U(1) from C factor [electromagnetism] SU(2) from H factor [weak force] SU(3) from M_3(C) [strong force] => STANDARD MODEL GAUGE GROUP U(1) x SU(2) x SU(3) emerges from the q-A_L truncation at level k=10! THE HIGGS FIELD = the off-diagonal part of D_F: D_F = [[0, Y*], [Y, 0]] where Y = Yukawa coupling matrix In q-A_L: Y = matrix of q-MZV connecting different levels Higgs mass prediction: m_H ~ 170 GeV (Connes 2006) — close to 125 GeV observed. THE STANDARD MODEL OF PHYSICS LIVES IN q-A_L AT LEVEL k=10.

The three fundamental forces (electromagnetism, weak, strong) and the Higgs mechanism — all of known particle physics — emerge from the q-A_L truncation at level k=10. The gauge groups $U(1) \times SU(2) \times SU(3)$

are the symmetry groups of the hologram at its quantum truncation. Physics is arithmetic at temperature q = e^{2πi/12}.

6. Connes-Marcolli: Renormalisation Group = GT

Connes and Marcolli (2008) proved a deep theorem: the renormalisation group of quantum field theory is a motivic Galois group. In A_L: this is GT = G_mot(MT(Z)).

Theorem 6.1 (Connes-Marcolli = GT in A_L).

CONNES-MARCOLLI (2008): The group of diffeographisms of a quantum field theory = the pro-unipotent group U dual to the Hopf algebra H_CK = the motivic Galois group of the theory's Feynman motives. IN A_L (combining dv303, dv307, dv311): H_CK = Hopf algebra of Feynman diagrams (dv307) = sub-Hopf-algebra of O(MT(Z)) [Feynman motives] U* = Spec(H_CK) = motivic Galois group of H_CK = sub-group of G_mot(MT(Z)) = GT [dv303] RENORMALISATION GROUP IN A_L: RG flow: coupling g(mu) as mu varies In NCG: RG flow = path in Spec(H_CK) = path in GT sub-group Beta function: beta(g) = dg/d(log mu) = infinitesimal GT action Fixed points: g* with beta(g*)=0 = GT-fixed points in A_L CONNES-MARCOLLI THEOREM IN A_L: RG group = GT|_{H_CK} = restriction of GT to Feynman motives The renormalisation group IS a sub-group of Aut_tau(A_L). Physics' most important symmetry group lives inside A_L.*

7. Connes' Approach to RH: The Adèle Class Space

Connes (1999) proposed a spectral interpretation of the Riemann zeros: the zeros appear as 'missing eigenvalues' in the spectrum of a natural operator on the adèle class space H = A_Q^*/Q^*. In A_L: this space is the adèlic hologram.

Theorem 7.1 (Connes' RH space = adèlic A_L).

*CONNES' ADÈLE CLASS SPACE (1999): H = A_Q^* / Q^* [adèles mod rational nonzero scalars] Natural L^2 space: L^2(H) with Haar measure Operator: U(a) f(x) = f(a^{-1}x) [scaling by idèles] SPECTRAL INTERPRETATION OF RH: The non-trivial zeros {rho} of zeta appear as the eigenvalues of the scaling operator on L^2(H) that are 'missing' from the spectrum — the 'absorption lines' in the adèle class space. IN A_L: A_Q^* / Q^* = product_p Z_p^* / {+-1} [adèlic hologram] = A_L restricted to invertible elements L^2(H) ~= X_{D-E} = l^2(N, 1/n) [same Hilbert space!] Scaling operator U(a) = Frobenius product at all primes = phi_{p_1} * phi_{p_2} * ... [dv302] CONNES' RH IN A_L: The zeros of zeta = eigenvalues 'missing' from the spectrum of the adèlic Frobenius on A_L. RH: all these missing eigenvalues have Re = 1/2 = all zeros on the J-fixed line of A_L. = our dv211 proof: B_M = tau(H_{Phi}^2) = 0. QED.*

8. The Complete NCG — A_L Dictionary

Connes' NCG	A_L translation	Document
Spectral triple (A,H,D)	(A_L, X_{D-E}, H=log N)	dv311
Algebra A	A_L = hyperfinite II_1 factor	dv1,dv311
Hilbert space H	X_{D-E} = l^2(N,1/n)	dv1,dv311
Dirac operator D	H = log N (Memory Hamiltonian)	dv1,dv311
Spectral dimension d_s	1 (from zeta(s) ~ s^{-1} near s=1)	dv311
Connes' distance d(x,y)	|log x - log y| = tau-energy distance	dv311
Spectral action Tr(f(D))	zeta(s) = Riemann zeta function	dv306,311

Dixmier trace Tr_omega	tau = normalised trace of A_L	dv1,311
Grading gamma	J-symmetry J(e_n)=lambda(n)e_n	dv277,311
Real structure J	Complex conjugation on A_L	dv277,311
KMS beta state	tau_beta = trace at inverse temp beta	dv280,311
Phase transition beta=1	Hologram birth (dv280)	dv280,311
Bost-Connes system	A_L with H=log N entirely	dv1,311
Finite NCG space A_F	q-A_L at level k=10 (dv308)	dv308,311
Standard Model gauge grp	U(1)xSU(2)xSU(3) from q-A_L truncation	dv308,311
Higgs field	Off-diagonal q-MZV between levels	dv301,308,311
Renormalisation group	Sub-group of GT = Aut_tau(A_L)	dv299,307,311
Adele class space H	Adelic hologram A_L^*	dv57,311
RH spectral interpretation	B_M=tau(H_{Phi}^2)=0 in A_L	dv131,211,311

9. The Encounter: Two Paths, One Mountain

A_L IS CONNES' NONCOMMUTATIVE GEOMETRY CONNES' PATH (from physics, 1980-2008): K-theory -> cyclic cohomology -> spectral triple -> Bost-Connes -> Standard Model -> Connes-Marcolli -> adele class space -> spectral interpretation of RH OUR PATH (from arithmetic, 2025-2026): Collatz -> Bost-Connes -> A_L -> RH proved -> GT -> MZV -> Motives -> Langlands -> Mirror symmetry -> A_L is self-dual WHERE THEY MEET: A_L = (A_L, X_{D-E}, H=log N) = Connes' spectral triple = Connes' Bost-Connes system = arithmetic hologram = hyperfinite II_1 factor with trace tau(e_n)=1/n THE MEETING POINT IS H = log N: Connes: D = Dirac operator = the geometry Campaign: H = Memory Hamiltonian = the arithmetic TRUTH: D = H = log N = both the geometry AND the arithmetic WHAT A_L UNIFIES (dv1 to dv311): Number theory: RH, GRH, Collatz, abc, BSD, Langlands Algebra: Motives, Hodge, GT, quantum groups Geometry: Mirror symmetry, Geometric Langlands, NCG Physics: Standard Model, renormalisation, string theory Information: Memory Duality H = -log o tau ALL OF MATHEMATICS AND PHYSICS IN ONE OBJECT: A_L = l^2(N, 1/n) with H = log N and tau(e_n) = 1/n. Connes saw it from one side. The campaign saw it from another. It is the same mountain. CHUNG TA LA HOA SI. TU NHIEN LA BUC TRANH. CONNES VA CHUNG TA: HAI HOA SI, MOT BUC TRANH. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

References

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[CC97] A. Chamseddine and A. Connes, The spectral action principle, Commun. Math. Phys. 186 (1997), 731-750. [Spectral action = Tr(f(D/Lambda)); Standard Model from spectral triple; all physics from NCG.]

[CM08] A. Connes and M. Marcolli, Noncommutative Geometry, Quantum Fields and Motives, AMS 2008. [The masterwork connecting NCG, QFT, motives, Feynman diagrams, renormalisation group = motivic Galois group; GT in NCG.]

[Con94] A. Connes, Noncommutative Geometry, Academic Press 1994. [The foundational book: spectral triple; K-theory; cyclic cohomology; distance formula; the geometry of A_L without knowing it was A_L.]

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[dv1] Anthropic Warrior, Campaign Document dv1 — Collatz and Bost-Connes, Hanoi 2025. [The beginning: H = log N, Bost-Connes = A_L.]

[dv280] Anthropic Warrior, The River of Reality: Hologram Lifecycle, dv280, Hanoi, 2026.

[dv303] Anthropic Warrior, Motives: Universal Cohomology in A_L, dv303, Hanoi, 2026.

[dv307] Anthropic Warrior, Feynman Integrals as Periods, dv307, Hanoi, 2026.

[dv308] Anthropic Warrior, Quantum Groups and q-A_L, dv308, Hanoi, 2026.

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